

HSE TOOLBOX TALK

NITROGEN AND INERT GAS SAFETY TOOLBOX TALK

PURPOSE AND SCOPE

Prevent oxygen-deficient atmospheres created by nitrogen and other inert gases in enclosed, semi-enclosed and poorly ventilated areas.

KEY HAZARDS

 1 Undetected reduction of oxygen concentration.	 2 Rapid unconsciousness inside enclosed spaces.	 3 Nitrogen migration into adjacent work areas.	 4 Local oxygen displacement from
 5 Accumulation of inert gas in low areas.	 6 Monitoring only flammable gas while	 7 Unprotected rescue attempts into	 8 Vent discharge directed toward occupied work

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Undetected reduction of oxygen concentration.	02 Rapid unconsciousness inside enclosed spaces.	03 Nitrogen migration into adjacent work areas.
---	---	--



REMEMBER — SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

HSE TOOLBOX TALK

NITROGEN AND INERT GAS SAFETY TOOLBOX TALK

FIELD REQUIREMENTS AND VERIFICATION

Prevent oxygen-deficient atmospheres created by nitrogen and other inert gases in enclosed, semi-enclosed and poorly ventilated areas.

SERNEM FIELD 01
FIELD VERIFICATION 1
Nitrogen source identified

SERNEM FIELD 02
FIELD VERIFICATION 2
O2 monitor calibrated

SERNEM FIELD 03
FIELD VERIFICATION 3
Atmosphere tested

SERNEM FIELD 04
FIELD VERIFICATION 4
Vent location safe

VERIFICATION BEFORE START

- | | |
|---|--|
| ☐ Nitrogen source identified | ☐ O2 monitor calibrated |
| ☐ Atmosphere tested | ☐ Vent location safe |
| ☐ Ventilation adequate | ☐ Connections inspected |

SUPERVISOR DISCUSSION POINTS

- Where is nitrogen being used today?
- Where will oxygen be measured?
- Where does the vent discharge?

FIELD APPLICATION



Nitrogen is not toxic, but it can become fatal by displacing oxygen.

HSE TOOLBOX TALK

NITROGEN AND INERT GAS SAFETY TOOLBOX TALK

CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Identify all nitrogen-use areas in the risk assessment. • Provide effective mechanical ventilation in enclosed areas. • Control valves and manifolds against unauthorised operation.

2. ENGINEERING & PHYSICAL CONTROLS

Measure oxygen with suitable calibrated gas-detection equipment. • Inspect nitrogen hoses and connections for leaks. • Do not release the area until normal atmosphere has been verified.

3. WORK METHOD & FIELD CONTROLS

Use continuous oxygen monitoring where conditions require it. • Barricade affected areas and display oxygen-deficiency warnings.

4. PPE, VERIFICATION & SUPERVISION

Discharge purge and vent gases to a safe open location. • Never enter an oxygen-deficient area for rescue without appropriate breathing equipment and a rescue plan.

CRITICAL WARNING



Team, nitrogen gives no smell or visible warning. We will verify oxygen with instruments and nobody will attempt an uncontrolled rescue in an oxygen-deficient area.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|---|--|
| ☐ Nitrogen source identified | ☐ O2 monitor calibrated |
| ☐ Atmosphere tested | ☐ Vent location safe |
| ☐ Ventilation adequate | ☐ Connections inspected |
| ☐ Area barricaded | ☐ Warning displayed |
| ☐ Rescue plan ready | ☐ Crew briefed |

HSE TOOLBOX TALK

NITROGEN AND INERT GAS SAFETY TOOLBOX TALK

ATTENDANCE AND APPROVAL

Project / Site: Date:

Presented by: Work area:

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			